

A Appendix

A.1 Proof of Proposition 1

Proof. Since $m^{(t+1)}$ minimizes $Q_n(\theta^{(t)}, m)$ over $\mathcal{M}$,

$$Q_n(\theta^{(t)}, m^{(t+1)}) \leq Q_n(\theta^{(t)}, m^{(t)}).$$

Similarly, since $\theta^{(t+1)}$ minimizes $Q_n(\theta, m^{(t+1)})$,

$$Q_n(\theta^{(t+1)}, m^{(t+1)}) \leq Q_n(\theta^{(t)}, m^{(t+1)}).$$

Combining the two inequalities gives

$$Q_n(\theta^{(t+1)}, m^{(t+1)}) \leq Q_n(\theta^{(t)}, m^{(t)}),$$

which proves the result. $\square$

A.2 Proof of Corollary 1

Proof. By Proposition 1, the sequence $\{Q_n(\theta^{(t)}, m^{(t)})\}_{t \geq 0}$ is nonincreasing, that is,

$$Q_n(\theta^{(t+1)}, m^{(t+1)}) \leq Q_n(\theta^{(t)}, m^{(t)}) \quad \text{for all } t \geq 0.$$

Moreover, since $Q_n(\theta, m)$ is a residual sum of squares,

$$Q_n(\theta, m) = \sum_{i=1}^n [Y_i - g(X_i; \theta) - m(Z_i)]^2 \geq 0,$$

for every admissible pair (θ, m) . Hence, $Q_n(\theta^{(t)}, m^{(t)}) \geq 0$, $t \geq 0$, showing that the sequence is bounded below. Therefore, $\{Q_n(\theta^{(t)}, m^{(t)})\}$ is a monotone decreasing sequence bounded from below. By the monotone convergence theorem for real sequences, there exists a finite constant $Q_n^* \geq 0$ such that

$$Q_n(\theta^{(t)}, m^{(t)}) \longrightarrow Q_n^* \quad \text{as } t \rightarrow \infty.$$

$\square$

A.3 Proof of Theorem 1

Proof. By compactness of Θ and boundedness of the objective sequence, the iterative sequence possesses at least one accumulation point. Let (θ^*, m^*) be such a point.

Since each iteration minimizes the objective with respect to one block of variables while keeping the other block fixed, continuity of $Q_n(\theta, m)$ implies that the limiting point

cannot be improved by varying only one block at a time. Hence

$$m^* = \arg \min_{m \in \mathcal{M}} Q_n(\theta^*, m) \quad \text{and} \quad \theta^* = \arg \min_{\theta \in \Theta} Q_n(\theta, m^*).$$

Therefore (θ^*, m^*) is a coordinatewise minimum and hence a stationary point of the alternating minimization procedure. $\square$

A.4 Proofs of Asymptotic Results

This appendix provides detailed proofs of the asymptotic results established in Section 4. The proofs are developed under the regularity assumptions (A1)–(A8) stated earlier. Since the model combines a finite-dimensional nonlinear parametric component and an infinite-dimensional isotonic regression component, the arguments involve both classical M-estimation theory and shape-constrained asymptotics. Throughout the discussion, let the true model be

$$Y_i = g(X_i; \theta_0) + m_0(Z_i) + \varepsilon_i, \quad i = 1, 2, \dots, n,$$

with $E(\varepsilon_i \mid X_i, Z_i) = 0$ and $E(\varepsilon_i^2 \mid X_i, Z_i) = \sigma^2 < \infty$. The least squares objective function is given by $Q_n(\theta, m) = \sum_{i=1}^n (Y_i - g(X_i; \theta) - m(Z_i))^2$. The joint estimator is obtained as $(\hat{\theta}, \hat{m}) = \arg \min_{\theta, m \in \mathcal{M}} Q_n(\theta, m)$.

A.5 Proof of Consistency, i.e., Theorem 2

Proof. To establish consistency of the estimator, consider the normalized least squares objective function

$$M_n(\theta, m) = \frac{1}{n} Q_n(\theta, m) = \frac{1}{n} \sum_{i=1}^n (Y_i - g(X_i; \theta) - m(Z_i))^2.$$

The idea is to show that this empirical criterion converges uniformly to its population counterpart and that the limiting objective has a unique minimizer at the true parameter values.

By the strong law of large numbers, for each fixed (θ, m) , $M_n(\theta, m) \xrightarrow{a.s.} M(\theta, m)$, where $M(\theta, m) = E[(Y - g(X; \theta) - m(Z))^2]$. Now substitute the true model representation as $Y = g(X; \theta_0) + m_0(Z) + \varepsilon$. Then, the population objective becomes

$$M(\theta, m) = E[\{g(X; \theta_0) - g(X; \theta) + m_0(Z) - m(Z) + \varepsilon\}^2].$$

Expanding the square gives

$$\begin{aligned} M(\theta, m) &= E \left[(g(X; \theta_0) - g(X; \theta))^2 \right] + E \left[(m_0(Z) - m(Z))^2 \right] \\ &\quad + 2E \left[(g(X; \theta_0) - g(X; \theta))(m_0(Z) - m(Z)) \right] + E(\varepsilon^2). \end{aligned}$$

The mixed terms involving the error disappear because of the conditional mean assumption $E(\varepsilon | X, Z) = 0$. Thus the criterion $M(\theta, m)$ is minimized when both components match their true counterparts, namely when $g(X; \theta) = g(X; \theta_0)$ and $m(Z) = m_0(Z)$.

By the identifiability assumption, this happens uniquely at $(\theta, m) = (\theta_0, m_0)$. Further, by standard uniform convergence arguments and compactness of the parameter space, the convergence of $M_n(\theta, m)$ to $M(\theta, m)$ is uniform over the admissible parameter space. Therefore, by the argmin consistency theorem, we conclude $(\hat{\theta}, \hat{m}) \xrightarrow{p} (\theta_0, m_0)$.

In particular, this implies $\hat{\theta} \xrightarrow{p} \theta_0$. □

A.6 Proof of Asymptotic Normality, i.e., Theorem 3

Proof. The proof is based on a standard Taylor expansion of the estimating equation around the true parameter value. Recall that the estimator $\hat{\theta}$ is obtained by solving the nonlinear estimating equation $S_n(\hat{\theta}, \hat{m}) = 0$, where

$$S_n(\theta, m) = \sum_{i=1}^n (Y_i - g(X_i; \theta) - m(Z_i)) \dot{g}(X_i; \theta), \text{ with } \dot{g}(X_i; \theta) = \frac{\partial g(X_i; \theta)}{\partial \theta}.$$

To derive the asymptotic distribution of $\hat{\theta}$, we expand $S_n(\hat{\theta}, \hat{m})$ around the true values (θ_0, m_0) . By a first-order Taylor expansion,

$$0 = S_n(\theta_0, m_0) + \left. \frac{\partial S_n(\theta, m_0)}{\partial \theta} \right|_{\theta=\theta_0} (\hat{\theta} - \theta_0) + \left. \frac{\partial S_n(\theta_0, m)}{\partial m} \right|_{m=m_0} (\hat{m} - m_0) + R_n.$$

where R_n denotes the remainder term.

Under the true values, the model becomes $Y_i = g(X_i; \theta_0) + m_0(Z_i) + \varepsilon_i$, so that $S_n(\theta_0, m_0) = \sum_{i=1}^n \varepsilon_i \dot{g}(X_i; \theta_0)$. Next, differentiating the score function with respect to θ , we obtain

$$\frac{\partial S_n(\theta, m_0)}{\partial \theta} = - \sum_{i=1}^n \dot{g}(X_i; \theta) \dot{g}(X_i; \theta)^\top + \sum_{i=1}^n (Y_i - g(X_i; \theta) - m_0(Z_i)) \ddot{g}(X_i; \theta),$$

where $\ddot{g}(X_i; \theta) = \frac{\partial^2 g(X_i; \theta)}{\partial \theta \partial \theta^\top}$. Evaluating at $\theta = \theta_0$, the second term has mean zero because $E(\varepsilon_i | X_i, Z_i) = 0$. Hence, by the law of large numbers, we can conclude that

$$\left. \frac{1}{n} \frac{\partial S_n(\theta, m_0)}{\partial \theta} \right|_{\theta=\theta_0} \xrightarrow{p} -A,$$

where $A = E [\dot{g}(X; \theta_0) \dot{g}(X; \theta_0)^\top]$. By assumption, A is positive definite.

Now consider the contribution from the isotonic component. Since the isotonic estimator converges to m_0 , its effect on the parametric estimator is asymptotically negligible at the root- n scale under the stated regularity conditions. Thus,

$$\left. \frac{\partial S_n(\theta_0, m)}{\partial m} \right|_{m=m_0} (\hat{m} - m_0) = o_p(\sqrt{n}).$$

Therefore, after dividing the Taylor expansion by $\sqrt{n}$, we obtain

$$0 = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i \dot{g}(X_i; \theta_0) - A \sqrt{n} (\hat{\theta} - \theta_0) + o_p(1),$$

that implies $\sqrt{n}(\hat{\theta} - \theta_0) = A^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i \dot{g}(X_i; \theta_0) + o_p(1)$. Now, by the multivariate central limit theorem, it is evident that $\frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i \dot{g}(X_i; \theta_0) \xrightarrow{d} N(0, B)$, where $B = E [\varepsilon^2 \dot{g}(X; \theta_0) \dot{g}(X; \theta_0)^\top]$. Finally, an application of Slutsky's theorem yields

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, A^{-1} B A^{-1}).$$

This establishes the asymptotic normality of the parametric estimator. □

A.7 Proof of Univariate Isotonic Limit, i.e., Theorem 4

Proof. The proof follows from the classical local asymptotic theory of isotonic regression. To study the local behavior of the isotonic estimator around a fixed point z , we introduce the local scaling $z_n = z + t n^{-1/3}$, where t is a fixed real number. This localization captures the characteristic cube-root asymptotic behavior of isotonic estimators.

Under the monotonicity constraint, the isotonic least squares estimator can be represented through the slope of the greatest convex minorant of an appropriate cumulative sum process. Specifically, consider the localized empirical process

$$U_n(t) = n^{2/3} [F_n(z + t n^{-1/3}) - F_n(z)],$$

where $F_n(\cdot)$ denotes the empirical cumulative process.

Using a second-order Taylor expansion of the underlying regression function $m_0(\cdot)$ around the point z , together with the local empirical process approximation, one obtains

$$U_n(t) = W_n(t) + \frac{1}{2}m'_0(z)t^2 + o_p(1),$$

where $W_n(t)$ converges weakly to a Gaussian process, namely $\sigma W(t)$, with $W(t)$ denoting standard Brownian motion.

The limiting behavior of the isotonic estimator is then obtained from the well-known switching relation, which connects the isotonic estimator with the slope of the greatest convex minorant of the limiting process. Consequently,

$$n^{1/3}(\widehat{m}(z) - m_0(z)) \Rightarrow \arg \min_{t \in \mathbb{R}} \{W(t) + t^2\}.$$

The random variable on the right-hand side is the classical Chernoff distribution, which arises naturally in shape-constrained inference. Therefore, after incorporating the appropriate scaling constant κ , we obtain that $n^{1/3}(\widehat{m}(z) - m_0(z)) \xrightarrow{d} \kappa \mathcal{C}$. □

A.8 Proof of Multivariate Isotonic Rate, i.e., Theorem 5

Proof. The proof is based on the complexity of the class of multivariate isotonic functions and standard empirical process arguments. Let $\mathcal{M}_q$ denote the collection of all coordinatewise monotone functions on $\mathbb{R}^q$. Since the estimator $\widehat{m}$ is obtained by minimizing the least squares criterion over this function class, its rate of convergence is governed by the size and geometric complexity of $\mathcal{M}_q$.

A fundamental result in shape-constrained estimation shows that the bracketing entropy of the isotonic class satisfies $\log N_{[]}(\epsilon, \mathcal{M}_q, L_2) \asymp \epsilon^{-q}$, which reflects the increasing complexity of the function class as the dimension q grows.

Using standard maximal inequalities from empirical process theory, the stochastic fluctuation of the empirical criterion around its expectation can be controlled uniformly over $\mathcal{M}_q$. In particular, for functions lying within an L_2 -neighborhood of radius δ around the true function m_0 , we have $\sup_{m \in \mathcal{M}_q} |(M_n - M)(m)| = O_p(n^{-1/2}\delta^{-q/2})$.

On the other hand, the deterministic part of the criterion behaves quadratically near the true function, contributing a term of order δ^2 . The rate of convergence is obtained by balancing these two quantities, namely by equating the stochastic fluctuation with the deterministic curvature as $\delta^2 \asymp n^{-1/2}\delta^{-q/2}$. By rearranging, we get $\delta^{q/2+2} \asymp n^{-1/2}$.

Solving for δ , we obtain $\delta \asymp n^{-1/(q+2)}$. This establishes the convergence rate of the isotonic estimator. Therefore, $\widehat{m}(z) - m_0(z) = O_p(n^{-1/(q+2)})$. Thus, the multivariate isotonic estimator converges at rate $n^{-1/(q+2)}$, which becomes slower as the dimension q increases, reflecting the well-known curse of dimensionality under shape constraints.

□

A.9 Proof of Theorem 6

Proof. The asymptotic linearization established in Theorem 3 implies that

$$\widehat{\theta} - \theta_0 = A^{-1} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \dot{g}(X_i; \theta_0) + o_p(n^{-1/2}), \quad (\text{A.1})$$

which proves the first assertion. For the isotonic component, Theorem 5 yields

$$\sup_z |\widehat{m}(z) - m_0(z)| = O_p(n^{-1/(q+2)}).$$

Next, using the definitions of the fitted and true residuals,

$$\widehat{\varepsilon}_i = Y_i - g(X_i; \widehat{\theta}) - \widehat{m}(Z_i), \quad \varepsilon_i = Y_i - g(X_i; \theta_0) - m_0(Z_i),$$

we obtain

$$\widehat{\varepsilon}_i - \varepsilon_i = -[g(X_i; \widehat{\theta}) - g(X_i; \theta_0)] - [\widehat{m}(Z_i) - m_0(Z_i)]. \quad (\text{A.2})$$

Applying a first-order Taylor expansion of $g(X_i; \widehat{\theta})$ around θ_0 gives

$$g(X_i; \widehat{\theta}) = g(X_i; \theta_0) + \dot{g}(X_i; \theta_0)^\top (\widehat{\theta} - \theta_0) + R_{ni},$$

where $R_{ni} = \frac{1}{2}(\widehat{\theta} - \theta_0)^\top \ddot{g}(X_i; \widetilde{\theta}_i)(\widehat{\theta} - \theta_0)$ for some intermediate point $\widetilde{\theta}_i$ between $\widehat{\theta}$ and θ_0 .

By assumptions (A3)–(A5) and the fact that $\widehat{\theta} - \theta_0 = O_p(n^{-1/2})$, it follows that $R_{ni} = O_p(n^{-1})$ uniformly in i , and therefore $\max_{1 \leq i \leq n} |R_{ni}| = o_p(n^{-1/2})$.

Substituting the Taylor expansion into (A.2) yields

$$\widehat{\varepsilon}_i - \varepsilon_i = -\dot{g}(X_i; \theta_0)^\top (\widehat{\theta} - \theta_0) - (\widehat{m}(Z_i) - m_0(Z_i)) + r_{ni}, \quad (\text{A.3})$$

where $r_{ni} = -R_{ni}$. Consequently, $\max_{1 \leq i \leq n} |r_{ni}| = o_p(n^{-1/2})$, which establishes the stated joint asymptotic representation.

□

A.10 Proof of Theorem 7

Proof. From Theorem 6, we have the residual representation

$$\widehat{\varepsilon}_i - \varepsilon_i = -\dot{g}(X_i; \theta_0)^\top (\widehat{\theta} - \theta_0) - (\widehat{m}(Z_i) - m_0(Z_i)) + r_{ni}, \quad (\text{A.4})$$

where $\max_{1 \leq i \leq n} |r_{ni}| = o_p(n^{-1/2})$. By Theorem 3, $\widehat{\theta} - \theta_0 = O_p(n^{-1/2})$, and Assumption (A4) implies $\max_{1 \leq i \leq n} \left| \dot{g}(X_i; \theta_0)^\top (\widehat{\theta} - \theta_0) \right| = O_p(n^{-1/2})$. Moreover, Theorem 5 yields $\sup_z |\widehat{m}(z) - m_0(z)| = O_p(n^{-1/(q+2)})$.

Since $\frac{1}{q+2} < \frac{1}{2}$, $q \geq 1$, it follows that $n^{-1/2} = o(n^{-1/(q+2)})$. Therefore, the isotonic estimation error dominates the parametric estimation error asymptotically. Taking maxima on both sides of (A.4) gives

$$\max_{1 \leq i \leq n} |\widehat{\varepsilon}_i - \varepsilon_i| \leq O_p(n^{-1/2}) + O_p(n^{-1/(q+2)}) + o_p(n^{-1/2}),$$

and hence $\max_{1 \leq i \leq n} |\widehat{\varepsilon}_i - \varepsilon_i| = O_p(n^{-1/(q+2)})$. This proves the first assertion.

To establish the second result, note that $(\widehat{\varepsilon}_i - \varepsilon_i)^2 = O_p(n^{-2/(q+2)})$, uniformly in i . Consequently, $\frac{1}{n} \sum_{i=1}^n (\widehat{\varepsilon}_i - \varepsilon_i)^2 \leq \max_{1 \leq i \leq n} (\widehat{\varepsilon}_i - \varepsilon_i)^2$. Using the first part of the theorem, we get $\max_{1 \leq i \leq n} (\widehat{\varepsilon}_i - \varepsilon_i)^2 = O_p(n^{-2/(q+2)})$, which implies $\frac{1}{n} \sum_{i=1}^n (\widehat{\varepsilon}_i - \varepsilon_i)^2 = O_p(n^{-2/(q+2)})$. $\square$

A.11 Proof of Theorem 8

Proof. Distance covariance and Bergsma–Dassios τ^* are continuous functionals of the joint empirical distribution of (W, ε) . By Theorem 7,

$$\max_i |\widehat{\varepsilon}_i - \varepsilon_i| \xrightarrow{p} 0.$$

A stochastic equicontinuity argument together with the Lipschitz continuity of the corresponding kernel functions yields

$$T_n(\widehat{\varepsilon}) - T_n(\varepsilon) = o_p(1).$$

The conclusion follows. $\square$